

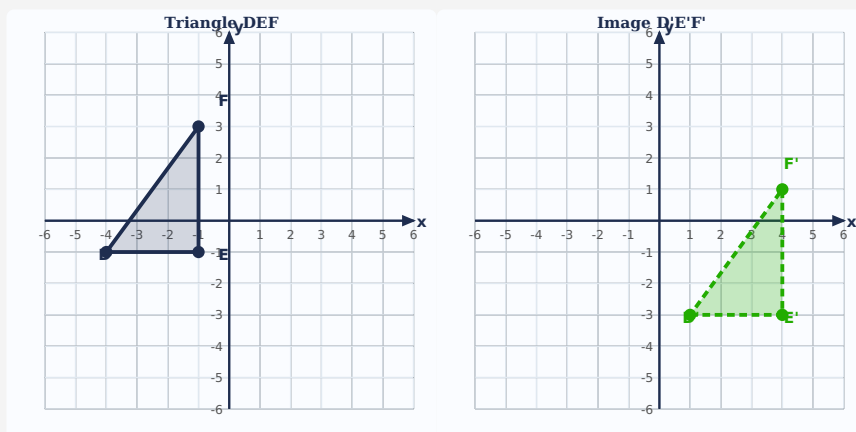
Unit 8 Homework — Transformations

Complete independently · NC.8.G.1-4 · Show all coordinate work

✓ ANSWER KEY — Unit 8 Homework (Transformations)

Translation: $(x, y) \rightarrow (x + h, y + k)$ — slides without rotating or flipping

Question 1 — Translate a Triangle



Triangle DEF was translated to $D'E'F'$. If $D = (-4, -1)$ and $D' = (1, -3)$, identify the translation rule.

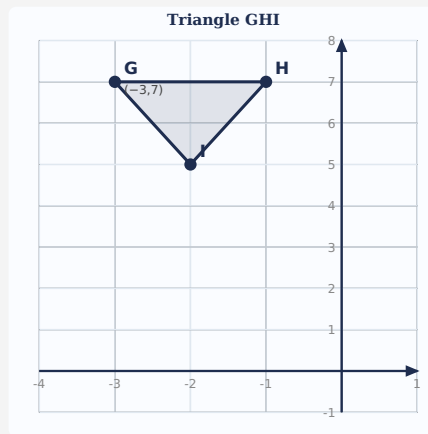
$(x - 5, y + 2)$

✓ $(x + 5, y - 2)$

$(x + 2, y - 5)$

$(x + 5, y + 2)$

Question 2



Triangle GHI is translated using $(x + 6, y - 5)$. Find G' ($G = (-3, 7)$).

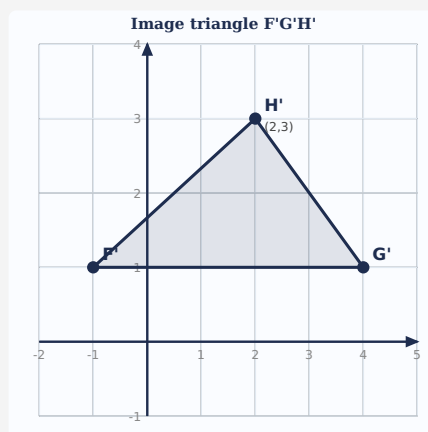
✓ **$G'(3, 2)$**

$G'(3, 12)$

$G'(-9, 12)$

$G'(3, -2)$

Question 3



Triangle F'G'H' is the image after the translation $(x - 4, y + 7)$. If $H' = (2, 3)$, find the original H.

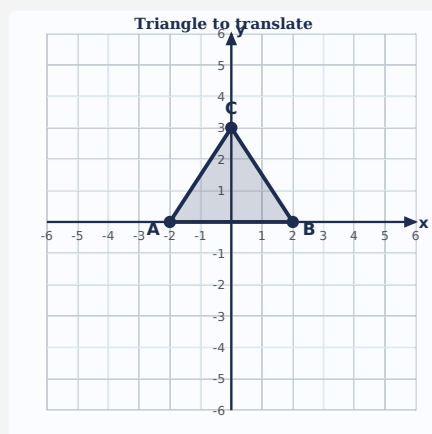
✓ **$H(6, -4)$**

$H(-2, 10)$

$H(-2, -4)$

$H(6, 10)$

Question 4



Triangle with vertices $(-2,0)$, $(2,0)$, $(0,3)$ is translated $(x+3, y-1)$. What is the new position of $(0,3)$?

✓ **$(3,2)$**

$(-2,2)$

$(3,0)$

$(0,2)$

Question 5

A figure is translated $(x+2, y-3)$, then translated again $(x-5, y+1)$. What single rule equals both translations combined?

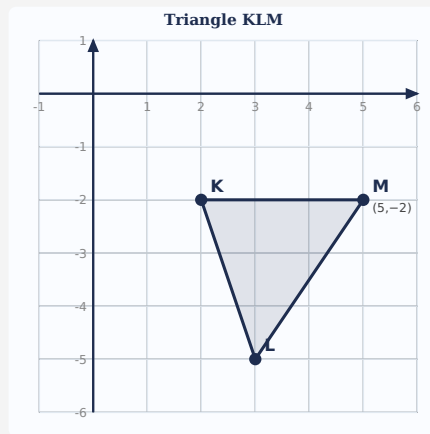
✓ **$(x-3, y-2)$**

$(x+3, y-2)$

$(x-7, y+4)$

$(x-3, y+2)$

Question 6



Triangle KLM is translated 3 units left and 6 units up. What are the coordinates of M'?

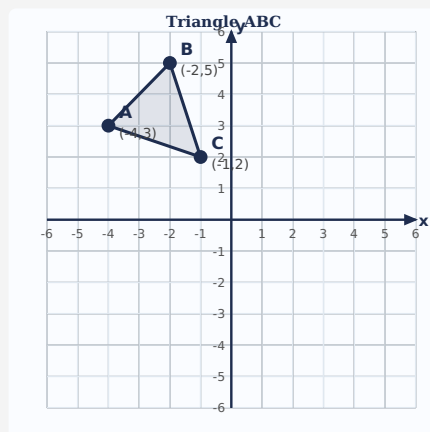
✓ M'(2, 4)

M'(8, 4)

M'(2, -8)

M'(8, -8)

Question 7 — Translate a Triangle



Triangle ABC is translated by $(x, y) \rightarrow (x+4, y-5)$. What are the coordinates of A'?

✓ A'(0, -2)

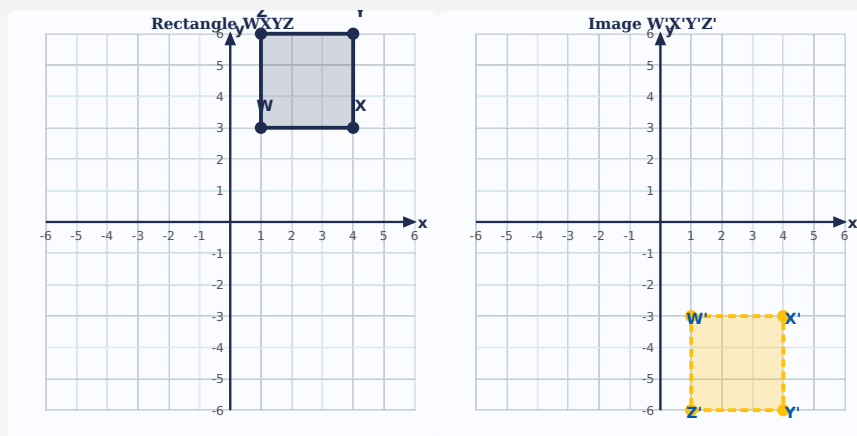
A'(-8, 8)

A'(0, 8)

A'(-2, 0)

x-axis reflection: $(x, y) \rightarrow (x, -y)$ | y-axis reflection: $(x, y) \rightarrow (-x, y)$

Question 1 — Reflect a Rectangle



Rectangle WXYZ was reflected. What line of reflection was used?

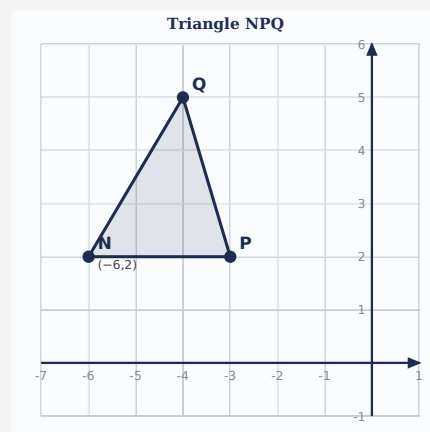
y-axis

✓ x-axis — y-coordinates changed sign, x stayed

$y = x$

$y = -1$

Question 2



Triangle NPQ is reflected over the y-axis. Find the image N'.

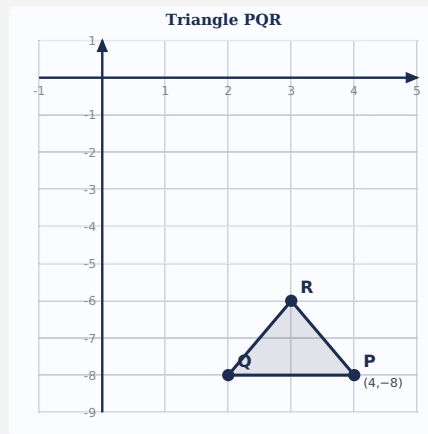
✓ N'(6, 2)

N'(-6, -2)

N'(6, -2)

N'(2, -6)

Question 3



Triangle PQR is reflected over the x-axis. Find P' ($P = (4, -8)$).

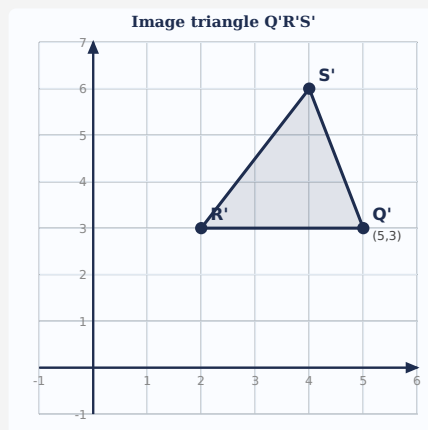
$P'(-4, 8)$

✓ $P'(4, 8)$

$P'(-4, -8)$

$P'(8, 4)$

Question 4



Triangle Q'R'S' is the image after a reflection over the y-axis. If $Q' = (5, 3)$, find the original Q.

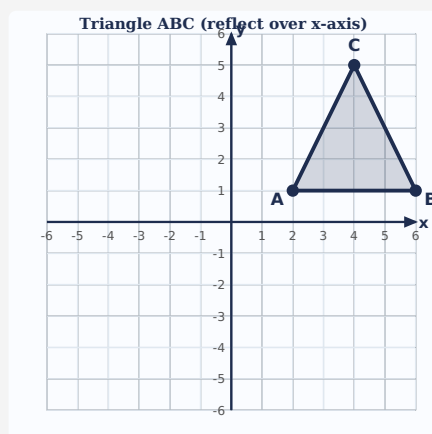
$Q(5, 3)$

✓ $Q(-5, 3)$

$Q(5, -3)$

$Q(3, 5)$

Question 5



Triangle A(2,1), B(6,1), C(4,5) reflected over x-axis. Find B'.

B'(-6, 1)

✓ B'(6, -1)

B'(-6, -1)

B'(1, 6)

Question 6

What do reflections over the x-axis and y-axis have in common?

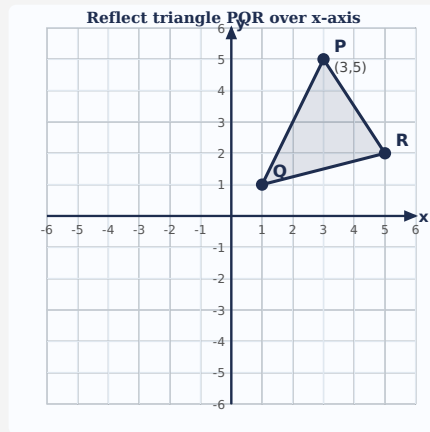
Both flip x-coordinates

✓ Both produce congruent (same size and shape) figures

Both always move figures to a new quadrant

Both are non-rigid transformations

Question 7 — Reflect a Triangle over the x-axis



Triangle PQR is reflected over the x-axis. What are the coordinates of P'?

✓ **P'(3, -5)**

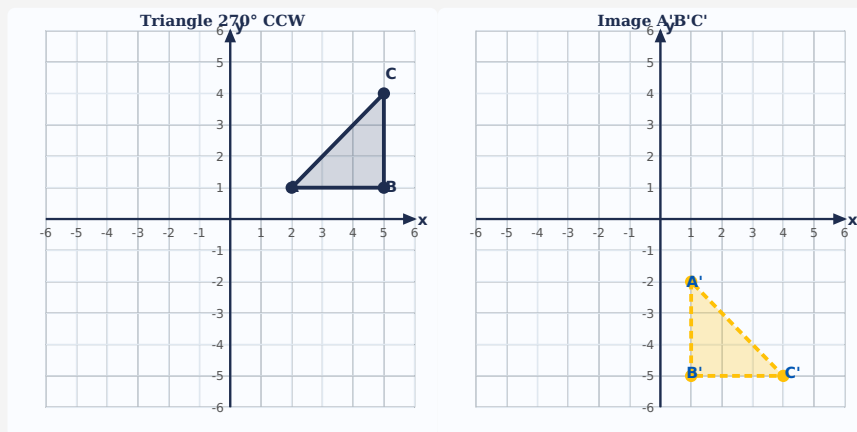
P'(-3, 5)

P'(-3, -5)

P'(5, 3)

$90^\circ \text{CCW}: (x, y) \rightarrow (-y, x)$ | $180^\circ: (x, y) \rightarrow (-x, -y)$ | $270^\circ \text{CCW}: (x, y) \rightarrow (y, -x)$ | $90^\circ \text{CW} = 270^\circ \text{CCW}$

Question 1 — 270° CCW Rotation



Triangle ABC was rotated 270° CCW. If $A = (2, 1)$, find A'.

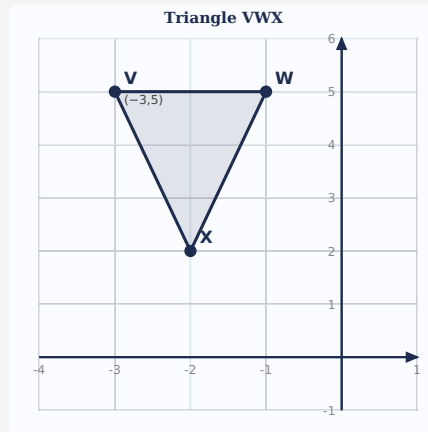
A'(-1, 2)

✓ **A'(1, -2)**

A'(-2, -1)

A'(2, -1)

Question 2



Triangle VWX is rotated 90° counterclockwise about the origin. Find V' .

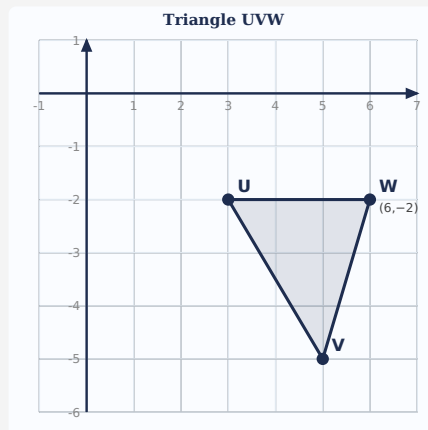
✓ $V'(-5, -3)$

$V'(5, 3)$

$V'(3, 5)$

$V'(-5, 3)$

Question 3



Triangle UVW is rotated 180° about the origin. Find W' .

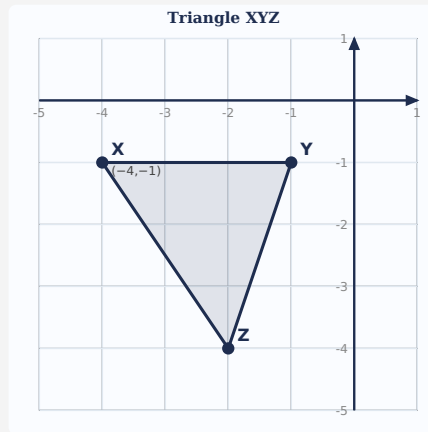
✓ $W'(-6, 2)$

$W'(2, 6)$

$W'(6, 2)$

$W'(-2, -6)$

Question 4



Triangle XYZ is rotated 90° clockwise ($= 270^\circ$ CCW) about the origin. Find X' .

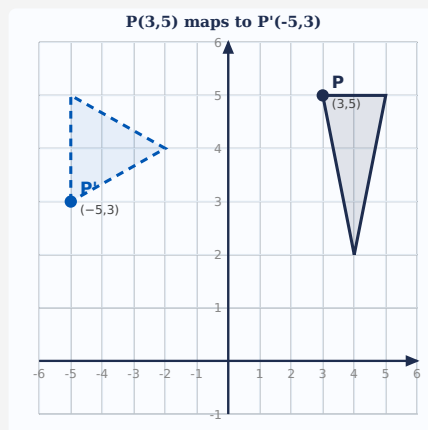
$X'(1, 4)$

$X'(4, 1)$

✓ $X'(-1, 4)$

$X'(1, -4)$

Question 5



Triangle PQR is rotated so that vertex $P(3, 5)$ maps to $P'(-5, 3)$. Which rotation was used?

90° CW

180°

✓ 90° CCW

270° CW

Question 6

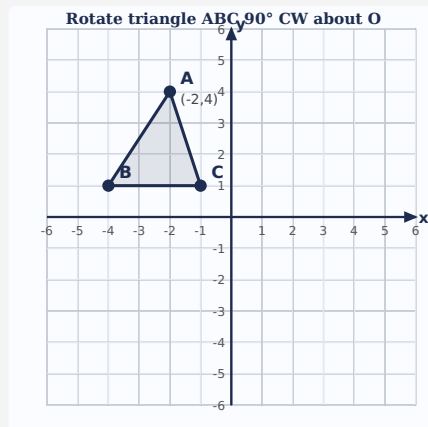
A figure is rotated 90° CCW three times. What single rotation gives the same result?

✓ **270° CCW**

90° CW

180°

360°

Question 7 — Rotate a Triangle 90° Clockwise

Triangle ABC is rotated 90° **clockwise** about the origin. What are the coordinates of A'?

✓ **$A'(4, 2)$**

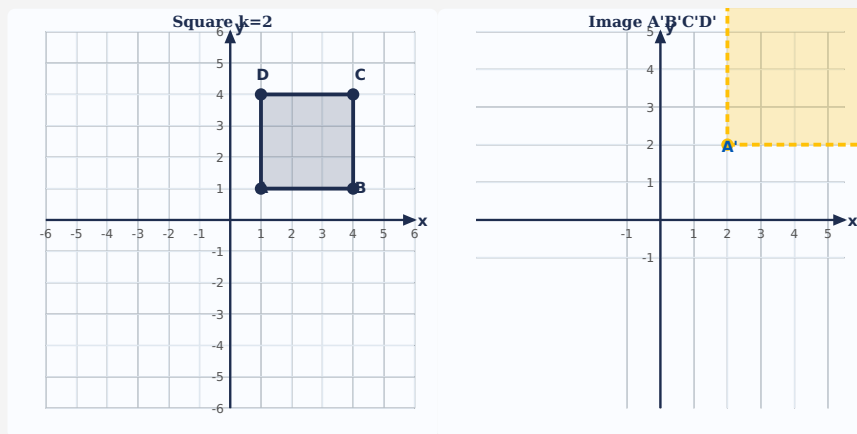
$A'(-4, -2)$

$A'(2, 4)$

$A'(-4, 2)$

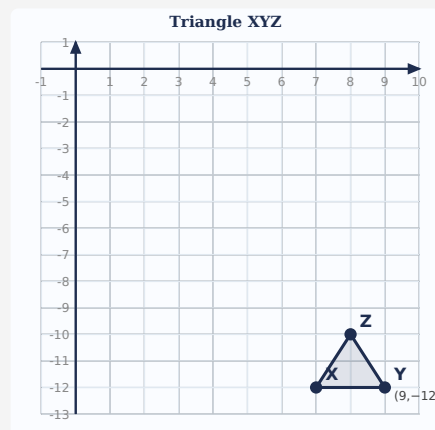
Dilation: $(x,y) \rightarrow (kx,ky)$ | $k > 1$ = enlargement | $0 < k < 1$ = reduction | Similar figures

Question 1 — Dilation $k = 2$



Square ABCD was dilated with $k = 2$. If $A = (1, 1)$, find A' .

Question 2



Triangle XYZ is dilated with scale factor $k = \frac{1}{3}$ about the origin. Find Y' ($Y = (9, -12)$).

Question 3

The image Z' is at $(-20, 8)$. If $k = 4$, find original Z .

✓ **$Z(-5, 2)$**

$Z(-80, 32)$

$Z(-5, -2)$

$Z(5, 2)$

Question 4

Rectangle with perimeter 20 is dilated $k = 3$. New perimeter?

✓ **60**

23

180

9

Question 5

Triangle area = 16 sq units. After dilation $k = \frac{1}{2}$, what is the new area?

8 sq units

✓ **4 sq units**

32 sq units

64 sq units

Question 6

A dilation maps every point farther from the origin. This is an:

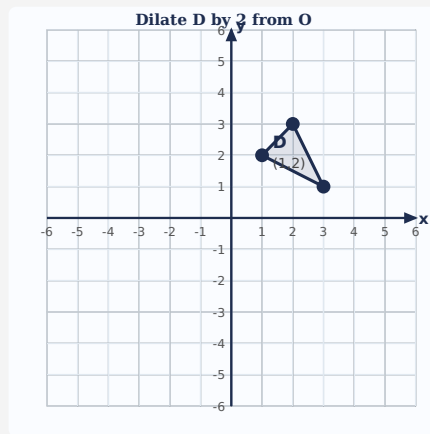
✓ **Enlargement ($k > 1$)**

Reduction ($0 < k < 1$)

Reflection

Translation

Question 7 — Dilation (scale factor 2)



Triangle DEF is dilated by a scale factor of 2 centered at the origin. What is the image of vertex D(1,2)?

✓ **D'(2, 4)**

D'(3, 4)

D'(1, 2)

D'(0.5, 1)

Mixed Review: These questions combine all transformation types.

Question 1

Which transformation changes the SIZE of a figure?

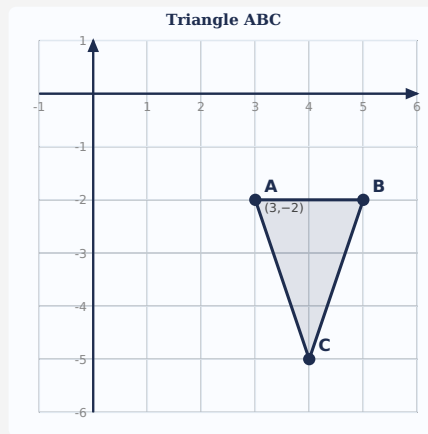
Translation

Reflection

Rotation

✓ **Dilation**

Question 2



Triangle ABC: apply a 90° CCW rotation, then a reflection over the x-axis. Find the final image A'' of $A(3, -2)$.

(2, 3)

✓ (2, -3)

(-2, 3)

(-2, -3)

Question 3

Figure M is translated, reflected, then rotated. Is the result congruent or similar to M?

Similar — three transformations change size

✓ Congruent — all three are rigid

Similar — rotation changes shape

Cannot determine

Question 4

What is the result of reflecting (a, b) first over the y-axis, then over the x-axis?

(a, b)

(-a, b)

(a, -b)

✓ (-a, -b)

Question 5

A 10-unit segment is dilated $k = 2.5$. What is the new length?

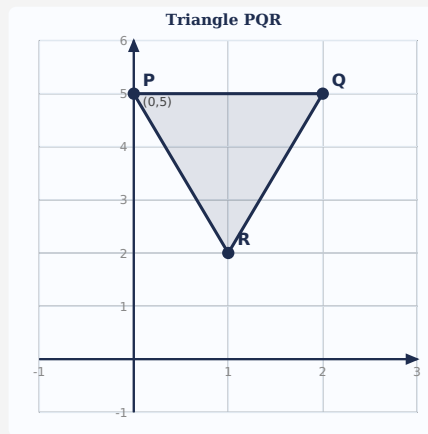
✓ 25 units

12.5 units

20 units

4 units

Question 6



Triangle PQR: translate by $(x + 3, y - 8)$, then dilate by $k = 2$ about the origin. Find P'' of $P(0, 5)$.

✓ **(6, -6)**

(3, -3)

(6, -16)

(3, -8)

Question 7

Figures X and Y are congruent. Which transformation was NOT used?

Translation

✓ **Dilation**

Reflection

Rotation

Question 8

What transformation maps EVERY point (x, y) to $(-x, -y)$?

Reflection over x-axis

Reflection over y-axis

✓ **Rotation 180° about origin**

Translation

Worked Examples

Study these step-by-step solutions before attempting the problems above.

Composite: translate $(2, 5)$ by $(x - 3, y + 1)$, then reflect over x-axis

Step 1: Translate: $(2 - 3, 5 + 1)$

$(-1, 6)$

Step 2: Reflect x-axis: $(x, -y)$

$(-1, -6)$

Step 3: Final answer

$(2, 5) \rightarrow (-1, 6) \rightarrow (-1, -6)$ ✓

🔍 Common Mistakes

Can you spot what went wrong?

Rotate $(3, -2)$ by 90° CCW

✗ **Student:** Student writes: $(-2, 3)$

✓ **Fix:** WRONG! Rule is $(-y, x)$. $y = -2$, so $-y = 2$. Correct: $(2, 3)$

💡 Be careful with negatives in rotation rules. $-(-2) = +2$.

Dilate $(4, 6)$ by $k = \frac{1}{2}$

✗ **Student:** Student writes: $(2, 8)$

✓ **Fix:** WRONG! $\frac{1}{2} \times 4 = 2$ ✓ but $\frac{1}{2} \times 6 = 3$, not 8. Correct: $(2, 3)$

💡 Multiply **BOTH** coordinates by the scale factor.

⚡ Key Tips for This Unit

- ✓ Translation: $(x+a, y+b)$ — slide. CONGRUENT.
- ✓ Reflection: x-axis $\rightarrow$ negate y. y-axis $\rightarrow$ negate x. CONGRUENT.
- ✓ Rotation 90° CCW: $(-y, x)$. 180° : $(-x, -y)$. 270° CCW: $(y, -x)$. CONGRUENT.
- ✓ Dilation: (kx, ky) . SIMILAR (not congruent unless $k=1$).

